

Bias Classifiers

Three classifiers, one interface:

SemanticClassifier

Embeds six exemplar sentences per bias class into centroid vectors at init (one-time cost), then cosine-sims every input against them. Fast at inference.

Accuracy lives or dies by exemplar quality, add more sentences to `_EXEMPLARS` to improve it. No LLM call per turn.

Hybrid classifier (default in middleware)

Runs both semantic and keyword layers, fuses them at 65/35 weighting, adds a confidence bonus when both layers agree. Degrades gracefully: if Ollama is unreachable it drops to keyword-only without crashing.

LLMClassifier

One extra Ollama call per user turn with `temperature=0.1`, parses structured JSON back. Slowest, most interpretable, the `last_reason` field tells you why it classified that way, which is useful for debugging the scaffold's decisions. Falls back to HybridClassifier on malformed JSON or timeout.

To swap classifier in the middleware, one line change:

```
from bias_classifier import LLMClassifier as _Classifier
```

To benchmark all three against your hardware:

```
python bias_classifier.py all
```

The test cases are in `_TEST_CASES` at the bottom

worth adding real inputs from the scaffold logs once you've run a few sessions, since exemplar/keyword coverage will have gaps specific to how interactions play out.